OUSSAMA BACCARA

Software Developer

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PROFESSIONAL SUMMARY

Student at INSAT, passionate about full-stack development, DevOps, systems, and cybersecurity. Experience gained through various personal projects, open-source contributions, and algorithmic practice. Seeking an opportunity to develop my skills within a professional team.

EDUCATION

2023 - 2025 **Integrated preparatory cycle – MPI**

National Institute of Applied Science and Technology (INSAT), Tunisia

2025 - 2026 **Engineering Degree in Industrial Computing and Automation**

National Institute of Applied Science and Technology (INSAT), Tunisia

EXPERIENCE

Open Source Contributions

- **aviralGarg05/NexumDB**
  + *DevOps & CI/CD* — **Designed** multi-stage Docker builds and **GitHub Actions pipelines** to **automate publishing** to Docker Hub on releases.
  + *Security & Optimization* — **Integrated** cargo audit for vulnerability scanning, added dependency files for reproducible builds, and **implemented** a dependency cache.

  Impact: PR **#**30, PR **#**32, PR **#**41 (and 2 others)

- **aopen-ug/conveyor**
  + *Kubernetes Orchestration* — **Built** a complete Helm chart for **deploying** Conveyor CI, **including** templates for Deployments, Services, ConfigMaps, and PersistentVolumeClaims.
  + *Automated Publishing* — **Improved** the helm-publish CI workflow to **automate** chart packaging and publishing to the *artifact registry***.**

  Impact: PR **#**161 (Feature: Helm Chart Packaging)

- **siri1404/llm-infrastructure**
  + *CI Security Pipeline* — **Architected** a robust CI pipeline **integrating** Bandit for security linting, pip-audit for vulnerability scanning, and Mypy/Black for type safety and formatting.
  + *Automated CD* — **Developed** a continuous deployment workflow via *GitHub Actions* to **build and push** containerized LLM services to the *GitHub Container Registry (GHCR).*

  Impact: PR **#**34

- **Open-J-Proxy/ojp**
  + *Distributed Transactions* — **Implemented** *XA (eXtended Architecture)* support for *Oracle Database* to ensure ACID compliance and data integrity across distributed environments.
  + *System Optimization* — **Improved** proxy reliability by making the maximum incoming message size configurable, preventing memory exhaustion.

  Impact: PR **#**149, PR **#**65

Also contributed to:

- *GCP PAM JIT* — **Refactored** CLI logic to support *IAM* requests at the Folder level alongside existing Project-level support. (PR **#**9)
- *Domino (Erlang)* — **Optimized** *GitHub Pages* deployment workflows via conditional execution and Pure Nix build **integration.** (PR **#**36)
- *LinkShrink* — **Improved** system observability by integrating *Prometheus metrics* for tracking redirect statistics. (PR **#**5)
- *Pariksha* — **Resolved** critical null reference errors in React frontend components. (PR **#**48)

ENGENEERING PROJECTS

⚙ Systems & Networking

- **HTTP-Over-TCP** · *Go*

A from-scratch replica of the HTTP/1.1 protocol built over raw TCP sockets — no frameworks or standard HTTP libraries. Implements request parsing, response formatting, and connection handling, mirroring real-world web server behavior at the protocol level.
Skills: TCP/IP · Protocol design · HTTP/1.1 · Low-level networking

- **RegEx-Engine** · **C++**

A regular expression engine built from the ground up — no standard regex library used. Parses regex strings to identify patterns and groupings, compiles the conditions, then evaluates user input against them. Demonstrates strong understanding of string parsing, state machines, and compiler fundamentals.
Skills: Parsing · State machines · Compiler design · Algorithms

☁ DEVOPS & INFRASTRUCTURE

- **Uniflow** · **Go**

A distributed Kafka-based log processing pipeline, deployed and orchestrated with Kubernetes and Helm charts. Designed for scalable, fault-tolerant ingestion and processing of high-volume log streams.
Skills: Apache Kafka · Kubernetes · Helm · Distributed systems · Java · Log processing

- **log-processing-system** · **Java**

A distributed Kafka-based log processing pipeline, deployed and orchestrated with Kubernetes and Helm charts. Designed for scalable, fault-tolerant ingestion and processing of high-volume log streams.
Skills: Apache Kafka · Kubernetes · Helm · Distributed systems · Java · Log processing

☼ Utilities & Tooling

- **dotfiles** · **Bash** & **Lua**

Full development environment configuration for Hyprland on Arch Linux. Includes custom setups for NeoVim, Tmux, and Bash, plus automation scripts and environment management tooling. Lua handles NeoVim plugin and editor optimization.
Skills: Bash scripting · Lua · NeoVim · Tmux · Arch Linux · Dev environment · Automation

⚠ SECURITY TOOLS

- **scanme** · **Python**

A network scanning tool built with Scapy, supporting multiple scan techniques including TCP, FIN, Window, and Stealth scans — enabling detailed network reconnaissance and host discovery.
Skills: Python · Scapy · Network scanning · TCP/IP · Penetration testing · Reconnaissance

- **EyeOnDirs** · **Python**

A URL directory enumeration tool for web reconnaissance, designed to discover hidden paths and endpoints on web servers.
Skills: Python · Web reconnaissance · Directory enumeration · Penetration testing

◈ Full-Stack Applications

- **MindGathering · MERN / Go+MySQL / Spring Boot+PostgreSQL**

The same full-stack web application rebuilt three times across different backend stacks — MERN, Go with MySQL, and Spring Boot with PostgreSQL — as a deliberate exercise in mastering different architectures and comparing their tradeoffs.
Skills: React · Node.js · MongoDB · Go · MySQL · Spring Boot · PostgreSQL · REST APIs · Full-stack · NGINX

- **InsideOut & InsideOut Website · Java · Thymeleaf · Javascript**

A compression and decompression tool paired with its public-facing website. Built with Java on the backend and Thymeleaf for server-side rendering, combining algorithmic work with a complete web interface.
Skills: Java · Thymeleaf · Compression algorithms · Server-side rendering · Full-stack

- **LockItOut · Java · MySQL**

A command-line tool for securely storing user credentials, with passwords encrypted before being saved to a database. Demonstrates practical understanding of encryption, secure storage, and credential management.
Skills: Java · CLI · Encryption · Password hashing · Database · Security

CERTIFICATES

Networking & Infrastructure

- **Networking Basics (Cisco)**
- **The Bits and Bytes of Computer Networking (Google)**
- **Intermediate Docker & Kubernetes (DataCamp)**

Systems & Linux

- **Technical Support Fundamentals (Google)**
- **Linux Commands and Shell Scripting (IBM)**
- **GitHub Foundations (DataCamp)**

Security & Programming

- **Penetration Testing & Forensics (IBM)**
- **Programming in Python (Coursera)**

LANGUAGES

- **English** — Fluent
- **Frensh** — Intermediate
- **German** — Beginner